A3-PS035-TC001

Evidence-Based Operator Network Fault Localisation

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Problem Description

Network faults disrupt critical services

Connectivity failures can affect businesses, essential services and everyday users, making fast diagnosis important.

Network data is fragmented

Logs, alarms, topology and performance metrics often sit in separate systems and must be reviewed together.

Manual troubleshooting takes time

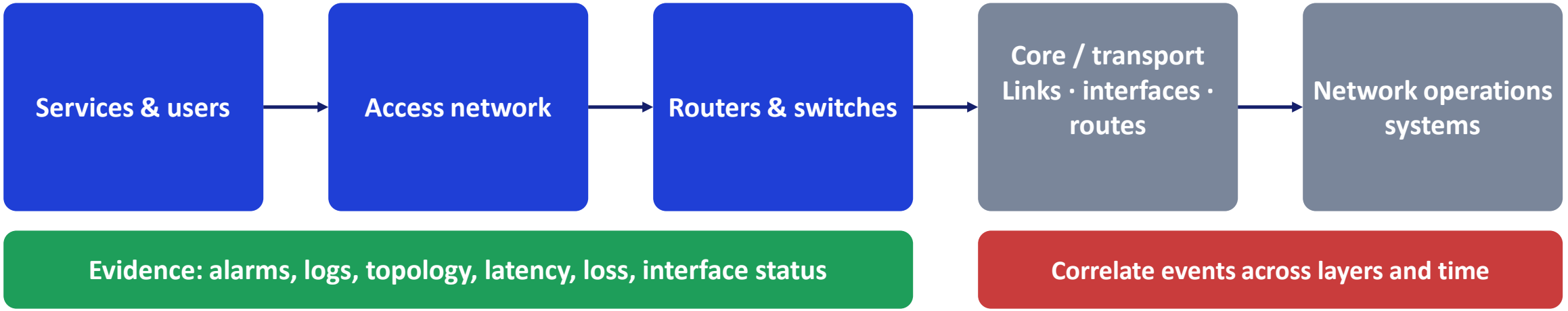
Operators may need to inspect large volumes of alerts and logs before narrowing down the likely fault source.

Operators need explainable results

A useful diagnosis should show the evidence behind a suspected fault rather than return an unsupported prediction.

Goal: correlate network events and performance evidence to identify probable fault locations and provide an auditable explanation to operators.

Background: From Network Signals to Fault Location



Multiple evidence sources

Logs, alarms and metrics together can help distinguish symptoms from likely root causes.

Timeline matters

Event order and time correlation can reveal which network change or failure came first.

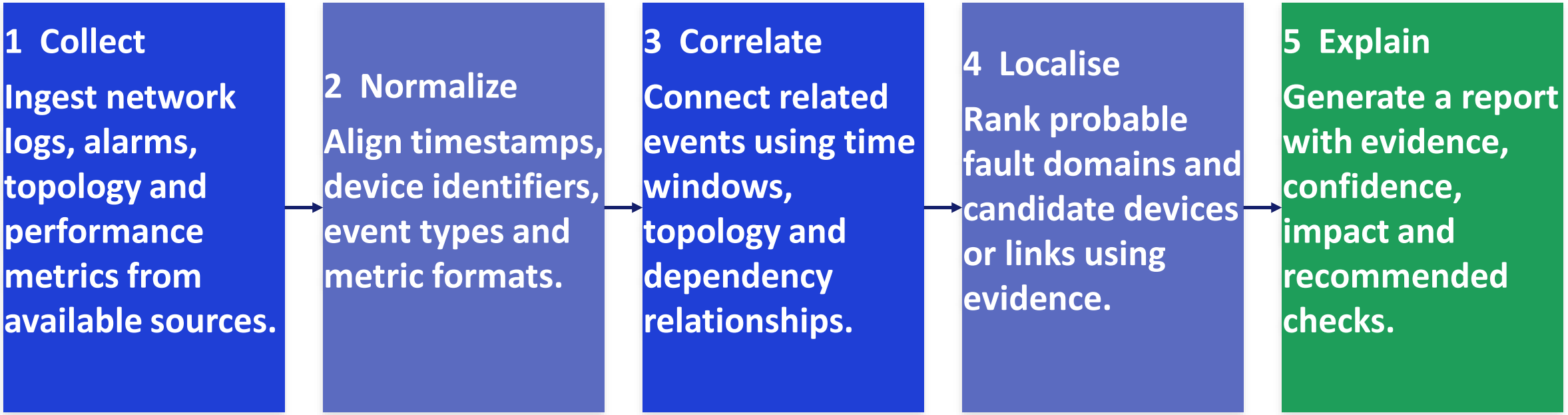
Evidence-first scope

Rank probable fault locations, expose supporting evidence and uncertainty, and leave final decisions to operators.

Related Work & Technical Foundations

Source / concept	Key idea	How we use it
Network monitoring	Collect counters, events and health metrics	Build a unified evidence view
Event correlation	Relate alarms by time, topology and dependency	Reduce duplicate alerts and connect symptoms
Topology-aware analysis	Use links and device relationships	Narrow down likely fault domains
Root-cause analysis	Rank candidate causes from observed symptoms	Produce explainable fault hypotheses
Explainable AI	Show factors supporting a prediction	Attach evidence and confidence to each result
Operator feedback	Human validation improves diagnosis quality	Capture confirmations and corrections

Proposed Solution: Collect, Correlate, Localise, Explain



Design principles

Evidence-linked results

Human review for high-impact decisions

Show uncertainty when evidence is incomplete

Protect sensitive network and operational data

Evidence-Based Fault Localisation Logic

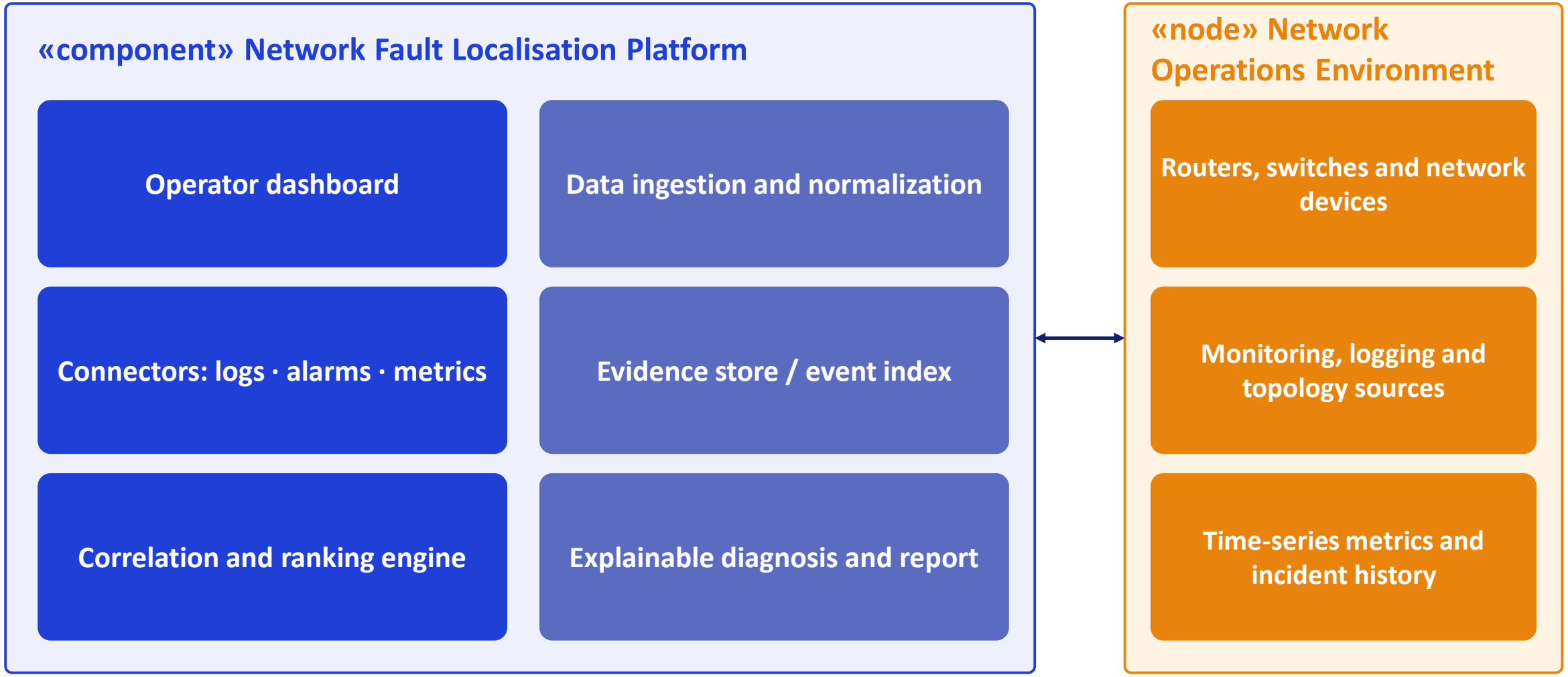
Stage	Evidence considered	Output
R1: Event timing	Related alarms and logs within a shared time window	Candidate incident group
R2: Topology	Affected devices, links and dependency paths	Likely fault domain
R3: Metrics	Interface errors, packet loss, latency and availability changes	Supporting or conflicting evidence
R4: Change context	Recent configuration or deployment events, if available	Possible triggering change
R5: Confidence check	Evidence agreement, gaps and contradictory signals	Ranked diagnosis or inconclusive result

Illustrative workflow: evidence availability and thresholds must be validated against the target network. The system reports probable causes, not guaranteed root causes.

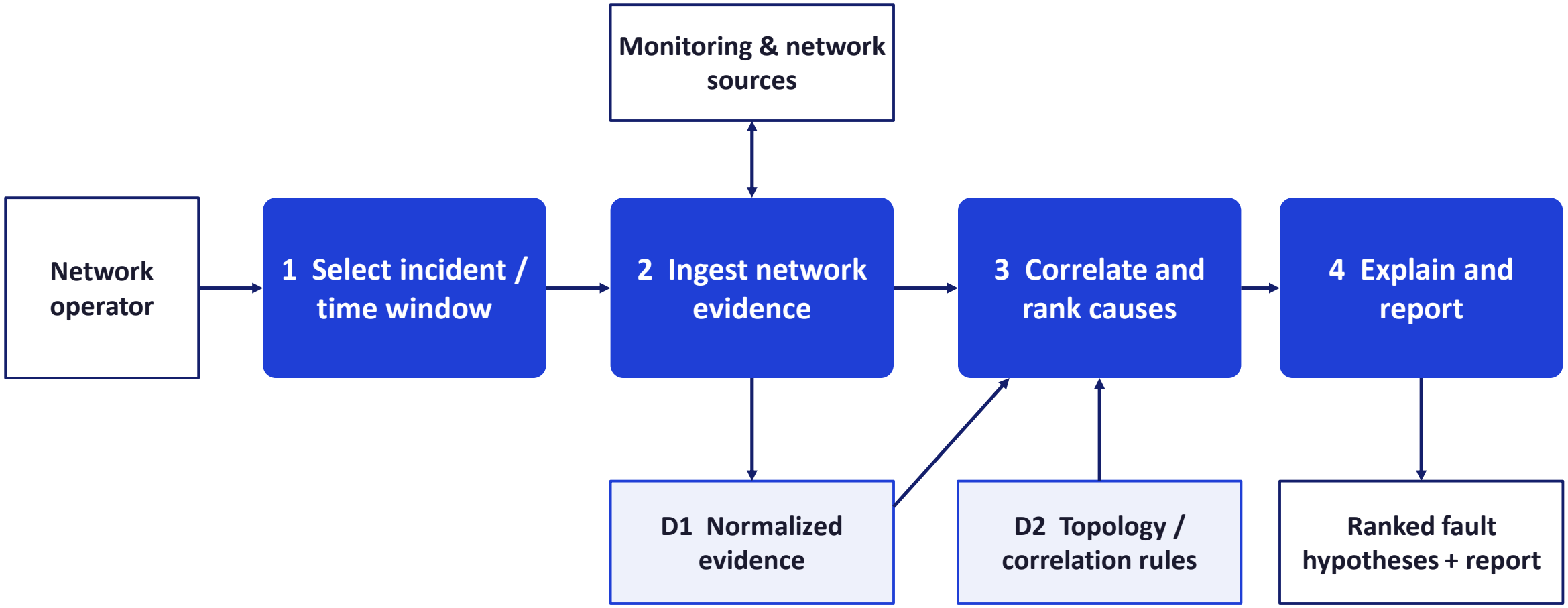
Optimization Proposed: Manual Triage vs Evidence-Driven Diagnosis

Aspect	Before: manual triage	After: proposed solution
Data review	Separate logs, alarms and metrics	Correlated evidence view
Fault search	Operator checks many possible causes	Ranked probable fault locations
Time context	Timeline assembled manually	Events aligned and correlated by time
Explainability	Reasoning may be difficult to reproduce	Evidence trail for each hypothesis
Uncertain cases	Repeated checks or guesswork	Explicit confidence and escalation

Solution Architecture & Design



Data Flow Diagram (DFD)



Inputs: monitoring systems and topology • Outputs: diagnosis, evidence trail and operator actions

Evaluation: Can the System Localise Faults Reliably?

Scenario	Test setup / evidence	Expected output	Evaluation measure
Link or interface failure	Interface-down event and correlated reachability change	Affected link / device ranked highly	Top-1 and Top-3 localisation accuracy
Packet loss / degradation	Loss and latency changes across related nodes	Likely affected segment	Correct fault-domain ranking
Configuration change	Change event followed by service degradation	Change listed as a candidate trigger	Evidence consistency
Multiple alarms	Burst of related alarms across topology	Correlated incident group	Alert reduction and correlation quality
Incomplete / conflicting data	Missing metrics or contradictory event signals	Low confidence / inconclusive	False-attribution rate

Evaluation targets should be measured on representative datasets or a controlled lab. Report localisation accuracy, time-to-diagnosis, false attribution and inconclusive cases; do not present targets as measured results.

Deployment Timeline

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Phase 1: Lab and probes

Build lab
ICMP, DNS, iperf3 probes
Poisson scheduler

Phase 2: Rules and faults

YAML rule engine
4 fault-injection scripts
Guided wizard

Phase 3: Evaluate

Repeated runs per fault
Inconclusive analysis
Tune thresholds

Method: build in small increments, validate each stage with labelled incidents where available, and document assumptions and limitations.

Method: private GitHub repo shared with mentors, small commits, a working increment demoed each phase.

References & Further Reading

[1] IETF, RFC 8632, A YANG Data Model for Alarm Management, 2019.

[2] IETF, RFC 8342, Network Management Datastore Architecture (NMDA), 2018.

[3] IETF, RFC 7950, The YANG 1.1 Data Modeling Language, 2016.

[4] IETF, RFC 9417, A YANG Data Model for Network Inventory, 2023.

[5] Research area: event correlation and root-cause analysis in communication networks.

[6] Research area: explainable AI for network operations and fault diagnosis.

[7] Project-specific datasets, topology records and monitoring documentation should be cited once selected.

Thank You for Your Attention!

Questions and discussion